

Power and Radical Axis

IMO Training Level 2

19/09/2020

1 Short Revision

1. Given a point P and a circle ω with radius R and centre O , the **power of point P with respect to ω** is defined to be

$$OP^2 - R^2$$

If a line through P intersects ω at X, Y , then

$$\text{power of } P \text{ w.r.t. } \omega = \begin{cases} PX \cdot PY, & P \text{ lies outside } \omega \\ -PX \cdot PY, & P \text{ lies inside } \omega \end{cases}$$

Also, if P lies outside ω , and PA is a tangent to ω (with A lying on ω), then power of $P = PA^2$.

2. Radical axis of two circles is the set of points which have same power w.r.t. the two circles. It is perpendicular to the line joining the centres of two circles.
3. Radical axis theorem.
4. **Remember you can always consider point circle (i.e. circle with zero radius).** In this case, the power of point P w.r.t. the point circle A is simply PA^2 .

2 Problems

1. (Euler's formula) Let O, I be the incentre and circumcentre of $\triangle ABC$. Suppose the circumcircle and incircle of $\triangle ABC$ have radii R, r respectively. Show that

$$OI^2 = R^2 - 2Rr$$

2. On the sides AB and AC of $\triangle ABC$ are given points P and Q , respectively, such that $PQ \parallel BC$. Prove that the circles with diameters BQ and CP intersect on the line through A that is perpendicular to BC .
3. D, E are points on sides AC and BC of $\triangle ABC$ such that $AD = BE$. Suppose the circumcircles of $\triangle ACE$ and $\triangle BCD$ intersect again at P . Show that CP bisects $\angle ACB$.
4. Let $\triangle ABC$ be a triangle with orthocentre H . Given a point P outside $\triangle ABC$. Suppose the line through H perpendicular to AP intersects line BC at X . Similarly define Y on AC and Z on AB . Show that X, Y, Z are collinear, and this line is perpendicular to PH .

5. Let ABC be a triangle, and let D, E, F be the points of contact of its incircle ω with its sides BC, CA, AB , respectively. Let K be the point of intersection of the line AD with the incircle ω different from D , and let M be the point of intersection of the line EF with the line perpendicular to AD passing through K . Prove that AM is parallel to BC .
6. In $\triangle ABC$, let D, E be the feet of altitudes from B and C respectively. Let M, I, K be the midpoints of BC, MD, ME respectively. Also, let N be a point on the perpendicular bisector of AM such that AN is parallel to BC . Show that K, I, N are collinear.
7. (ELMO 2014 G2) $ABCD$ is a cyclic quadrilateral inscribed in the circle ω . Let $AB \cap CD = E$, $AD \cap BC = F$. Let ω_1, ω_2 be the circumcircles of AEF, CEF , respectively. Let $\omega \cap \omega_1 = G$, $\omega \cap \omega_2 = H$. Show that AC, BD, GH are concurrent.
8. (HKTST 2018 Test 2 P4) In triangle ABC with incentre I , let M_A, M_B and M_C be the midpoints of BC, CA and AB respectively, and H_A, H_B and H_C be the feet of the altitudes from A, B and C to the respective sides. Denote by ℓ_b the line being tangent to the circumcircle of triangle ABC and passing through B , and denote by ℓ'_b the reflection of ℓ_b in BI . Let P_B be the intersection of $M_A M_C$ and ℓ_b , and let Q_B be the intersection of $H_A H_C$ and ℓ'_b . Define $\ell_c, \ell'_c, P_C, Q_C$ analogously. If R is the intersection of $P_B Q_B$ and $P_C Q_C$, prove that $RB = RC$.
9. In $\triangle ABC$ with centroid G , M and N are the midpoints of $\overline{AB}$ and $\overline{AC}$, and the tangents from M and N to the circumcircle of $\triangle AMN$ meet $\overline{BC}$ at R and S , respectively. Point X lies on side $\overline{BC}$ satisfying $\angle CAG = \angle BAX$. Show that $\overline{GX}$ is the radical axis of the circumcircles of $\triangle BMS$ and $\triangle CNR$.
10. (IMO 2014 G3) Let Ω and O be the circumcircle and circumcentre of an acute-angled triangle ABC with $AB > BC$. The angle bisector of $\angle ABC$ intersects Ω at $M \neq B$. Let Γ be the circle with diameter BM . The angle bisectors of $\angle AOB$ and $\angle BOC$ intersect Γ at points P and Q , respectively. The point R is chosen on the line PQ so that $BR = MR$. Prove that BR is parallel to AC .
11. (IMO 2009 G4) Given a cyclic quadrilateral $ABCD$, let the diagonals AC and BD meet at E and the lines AD and BC meet at F . The midpoints of AB and CD are G and H , respectively. Show that EF is tangent at E to the circle through the points E, G and H .
12. In acute scalene $\triangle ABC$ with incenter I , M is the midpoint of $\overline{BC}$ and L is the midpoint of arc $\widehat{BAC}$. Lines through M parallel to $\overline{BI}$ and $\overline{CI}$ meet $\overline{AB}$ and $\overline{AC}$ at E and F , respectively. $\overline{LB}$ and $\overline{LC}$ meet the circumcircles of $\triangle BME$ and $\triangle CMF$ again at P and Q , respectively. Show that I lies on the radical axis of the circumcircles of $\triangle PMQ$ and $\triangle EMF$.